The glue exponent

$d(k) \cdot (2^k + 1) = 2^{3k} + 1$, read slowly

Abstract

One arithmetic identity.

- `d_mul_gold`: $d(k) \cdot (2^k + 1) = 2^{3k} + 1$, where $d(k) = 2^{2k} - 2^k + 1$.

Two lines of `zify`/`ring`, but it is the algebraic glue between the *Kasami* exponent $d(k)$ and the *Gold* exponent $2^k + 1$, and it underlies "decompose Φ as a composition of two bijections". The cross-form factorization plays the same connective role.

Two exponents

- **Gold:** $2^k + 1$.
- **Kasami:** $d(k) = 2^{2k} - 2^k + 1$.

They look unrelated. The identity says they multiply to a clean cube:

$$d(k) \cdot (2^k + 1) \;=\; 2^{3k} + 1.$$

1 The lemma

```
theorem d_mul_gold (k : ℕ) (hk : k ≥1) :
    d k * (2 ^ k + 1) = 2 ^ (3 * k) + 1 := by
  unfold d
  zify
  rw [Nat.cast_sub (by gcongr <;> linarith)]
  push_cast
  ring
```

`d` is defined with *truncated* natural subtraction ($2^{2k} - 2^k$), so the proof first lifts to $\mathbb{Z}$ with `zify` and `Nat.cast_sub`, where subtraction behaves, then finishes with `ring`.

Why it is true (the cube identity)

Over $\mathbb{Z}$, set $q = 2^k$. Then $d(k) = q^2 - q + 1$ and the Gold factor is $q + 1$. This is the classical sum-of-cubes factoring:

$$(q+1)(q^2 - q + 1) \;=\; q^3 + 1 \;=\; 2^{3k} + 1.$$

$$\underbrace{2^k + 1}_{\text{Gold}} \times \underbrace{2^{2k} - 2^k + 1}_{\text{Kasami } d(k)} \xlongequal{q^3+1} 2^{3k} + 1$$

$x^3 + 1 = (x+1)(x^2 - x + 1)$ is the reason the Gold exponent and the Kasami exponent are "complementary cofactors" of the cube $2^{3k} + 1$.

2 Where it lands: composing two bijections

On $\mathrm{GF}(2^n)$, raising to a power is a bijection iff the exponent is coprime to $2^n - 1$. The identity factors the Kasami power map through Gold:

$$\left(x^{\,d(k)}\right)^{\,2^k+1} \;=\; x^{\,d(k)\,(2^k+1)} \;=\; x^{\,2^{3k}+1}.$$

So the cube power $x \mapsto x^{2^{3k}+1}$ *is* Gold-after-Kasami. When both right-hand factors are permutations, the identity lets the analysis *decompose* the differential as a composition of two bijections — the strategy behind the Dempwolff–Müller route to Kasami APN.

The same role: cross-form factorization

The companion connective identity is the cross form. With $L_k(P) = P^{2^k} + P$ and $N_k(s) = s^{2^k+1}$:

$$\mathrm{Cross}_k(s, P) \;=\; s\,P^{2^k} + s^{2^k} P \;=\; N_k(s) \cdot L_k\big(P/s\big).$$

```
def L (k : ℕ) (x : F) : F := x ^ (2 ^ k) + x
def N (k : ℕ) (x : F) : F := x ^ (2 ^ k + 1)
def Cross (k : ℕ) (s P : F) : F := s * P ^ (2 ^ k) + s ^ (2 ^ k) * P
```

Just as `d_mul_gold` factors an *exponent* as Gold $\times$ Kasami, the cross form factors a *value* as Gold-norm $N_k(s)$ times truncated-trace $L_k(P/s)$. Both turn one hard object into a product of two understood ones.

Pattern: a one-line algebraic factoring (of an exponent, or of a value) turns "analyse Φ" into "analyse a product of two bijections". The cube identity $x^3+1=(x+1)(x^2-x+1)$ and the cross factorization are the two faces of this glue.

One-screen summary

object	**factoring**
exponent	$d(k)\,(2^k+1)=2^{3k}+1 \;\; (q^3+1=(q+1)(q^2-q+1))$
power map	$x^{2^{3k}+1}=(x^{d(k)})^{2^k+1}$ (Gold $\circ$ Kasami)
value	$\mathrm{Cross}_k(s,P)=N_k(s)\cdot L_k(P/s)$

$d(k)\,(2^k+1)=2^{3k}+1 \;\Rightarrow\;$ Kasami power = Gold $\circ$ Kasami $\;\Rightarrow\;$ "compose two bijections".